

V1: Forward-Model Mismatch

9 October 2026 (rerun)

What parameter bias does forward-model error cause?

Amir Dollie (DLLAMI004), EEE4022S

SHARC Code: `runV1ModelMismatch.m`, `v1Tools.m`, `diagnoseV1Truncation.m`, commit `ee19555`

Results: `Results/V1/V1_20261009_105051`, `diagnostics/truncationLadder_20261009_101105`

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 4 Oct run were low by $\alpha(\omega) = H\omega^2/g$; `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). This summary supersedes `V1_20261004_121316`.

At p_0 the uncertainty fell about sevenfold, so F grew about fiftyfold, and every model discrepancy now measures about seven times larger in d . Two consequences:

- The stage 0 ladder no longer meets its rule. The truth truncation is `[210 20 20]`, the best available and **not converged** (logged 2026-10-09), so V1a is a lower bound.
- Water depth is no longer benign in Fisher units.

All registered settings are unchanged; tester 11/11.

Summary

Question. Every study so far generated and inverted with the same EMM (the “inverse crime”). M1 showed that the estimator behaves as predicted when the model is right. V1 generates the data from a *perturbed* truth model and keeps the frozen nominal inverse, to separate forward-model error from estimator uncertainty.

Outcome. The inverse is now precise enough ($\sigma_{\ln \beta} = 0.050\%$) that every tested model error is statistically significant. Measured against a physical 1% stiffness criterion, only sensor geometry matters.

- **Truncation** (production `[50 10 10]` against `[210 20 20]`): $d_{\text{sys}} = 10.0$ to 29.1 at the four floes, which makes every floe Not identifiable. The physical shifts are at most 0.18% in β and 0.12% in R . This is a lower bound, since the truth is not converged.
- **Water depth:** $d_{\text{sys}} = 0.5$ at $\pm 2.4\%$ and 1 at $\pm 4.8\%$, but $|\delta\beta/\beta| \leq 0.021\%$ at $\pm 5\%$. The 1% tolerance is not reached.
- **Common radial sensor error:** $d_{\text{sys}} = 0.5$ at $8.9 \mu\text{m}$ and 1 at $17.9 \mu\text{m}$; $|\delta\beta/\beta| = 1\%$ at 0.44 mm . Centimetre errors are far outside the linear regime (confirmed by a nonlinear inversion).

1 What the study does

Why V1 exists. A real floe is never exactly the EMM: the series is truncated, the depth is measured with error and the sensors are not exactly where the model places them. With $\sigma_{\ln \beta} = 0.050\%$ and $\sigma_{\ln R} = 0.0083\%$ at p_0 , even a tiny model-induced shift can be statistically decisive. V1 measures it in units of that uncertainty and in physical units.

Method (pre-registered in `cfg.v1`; clarifications and one pre-run correction logged before the 4 Oct run; the truth truncation logged 2026-10-09).

1. **Systematic bias.** $b_{\text{sys}} = f_W(\text{truth}) - f_W(\text{nominal})$: both are the Welch-expected features from one function, projected with the *nominal* J , Σ and F :

$$\delta\theta_{\text{sys}} = F^{-1} J^T \Sigma^{-1} b_{\text{sys}}, \quad d_{\text{sys}} = \sqrt{\delta\theta_{\text{sys}}^T F \delta\theta_{\text{sys}}}.$$

The class uses d_{tot} from the *vector* sum $\delta\theta_W + \delta\theta_{\text{sys}}$.

2. **V1a truncation.** Stage 0 selects the truth truncation by a ladder at p_0 ($M = 90, 110, \dots, 210$ with $[P N] = [20 20]$): the first step that moves the features by $d < 0.1$. If none does, the last level is the best available truth and is not converged. The inverse `[50 10 10]` is compared against that truth at p_0 and at the three large- kR A1 floes.
3. **V1b water depth.** True $H = H_0(1 + \epsilon_H)$ with $\epsilon_H \in \{\pm 2, \pm 5\}\%$, physically consistent ($\beta \propto H^{-4}$; $\gamma, R, r_i \propto H^{-1}$), with the physical sea and frequency band fixed.
4. **V1c radial sensor position.** Every sensor ($r > 0$) is moved by a common signed $\Delta r \in \{\pm 10, \pm 20\} \text{ mm}$, with angles unchanged.
5. **Tolerances.** The first crossing per side of $d_{\text{sys}} = 0.5$, $d_{\text{sys}} = 1$ and $|\delta\beta/\beta| = 1\%$, bracketed by the registered points and bisected in $\log_{10} |\text{error}|$ to 0.005 decade. Not extrapolated beyond the registered range.
6. **Nonlinear check** (diagnostic). The largest- d_{sys} case is inverted with G3’s solver from θ_{true} on its noiseless Welch features, and compared with $\theta_{\text{true}} + \delta\theta_W + \delta\theta_{\text{sys}}$.

Depth and radial cases are at p_0 (Level 2C, s26, LSM6DSV16X) and use the inverse truncation. The nominal Welch features reproduce the engine’s $f_0 + b_W$ exactly, so b_{sys} has no implementation offset. A clean tree was enforced (commit `ee19555`, 65 min, of which 17 min was the nonlinear check).

2 Results

2.1 Stage 0: truncation ladder

M step ($[P N] = [20 20]$)	90→110	110→130	130→150	150→170	170→190	190→210
d at p_0	1.154	0.662	0.410	0.264	0.184	0.151
$\delta \ln \beta$ per step	−0.003%	−0.010%	−0.010%	−0.009%	−0.008%	−0.008%

No step falls below $d = 0.1$, so [210 20 20] is the best available truth and is **not converged**. The diagnostics agree:

- The last step at the large- kR floes gives $d = 0.094$, 0.134 and 0.056, so the $\beta \times 0.5$, $R \times 2$ floe is not converged either.
- Raising P to 30 at $M = 210$ gives $d = 0.168$, so P is not converged.

The steps decay slowly (last ratio 0.82), each still moving β by about 0.008%, a sixth of $\sigma_{\ln \beta}$. If they keep shrinking geometrically, the remaining tail at p_0 is of order $d \approx 1$. The V1a distances are therefore lower bounds, uncertain by roughly 10% at p_0 . On 4 Oct the same rule selected [130 20 20]; the rule has not changed, but F has.

2.2 V1a: truncation bias

Floe	d_{sys}	$\delta\beta/\beta$	$\delta R/R$	d_W	d_{tot}	Class: matched $\rightarrow$ mismatched
p_0	9.98	-0.180%	+0.029%	0.436	10.01	Identifiable $\rightarrow$ Not identifiable
$R \times 2$	25.05	+0.037%	+0.101%	0.665	25.67	Marginal $\rightarrow$ Not identifiable
$\beta \times 0.5$, $R \times 2$	29.13	+0.125%	+0.123%	1.065	30.19	Not identifiable $\rightarrow$ Not identifiable
$\beta \times 2$, $R \times 2$	10.32	+0.068%	+0.051%	0.289	10.49	Identifiable $\rightarrow$ Not identifiable

Lower bounds (truth not converged). Pre-amendment (against [130 20 20]): d_{sys} 0.77 to 3.22, p_0 -0.031% in β .

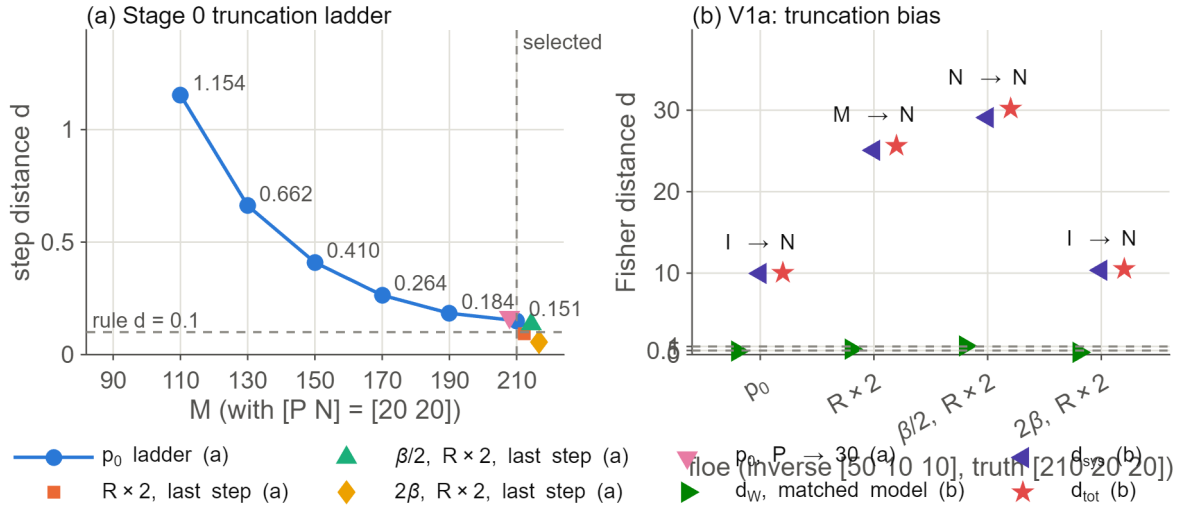


Figure 1. (a) Stage 0: feature change of each ladder step at p_0 (line) against the $d = 0.1$ rule, with the last step [190 $\rightarrow$ 210] at the large- kR floes and the $P \rightarrow 30$ check (markers offset slightly in M). (b) V1a per floe: the Welch bias of the matched model d_W , the truncation bias d_{sys} and their vector total d_{tot} , with the class change (I, M, N). The class limits $d = 0.5$ and 1 lie at the bottom of the axis.

The truncation bias exceeds the Welch bias by a factor 23 to 38. In physical terms the production model is accurate: no floe moves by more than 0.18% in β or 0.12% in R . But σ is now so small that these shifts are 10 to 30 uncertainty units. The p_0 stiffness shift is larger than on 4 Oct (-0.180% against -0.031%) for two reasons: the truth is further converged, and the amendment reweighted the bins.

2.3 V1b and V1c: depth and radial position

Case	d_{sys}	$\delta\beta/\beta$	$\delta R/R$	d_{tot}	Class
$\epsilon_H = -5\%$	1.026	-0.021%	+0.003%	1.114	Not identifiable
$\epsilon_H = -2\%$	0.413	-0.008%	+0.001%	0.603	Marginal
$\epsilon_H = +2\%$	0.417	+0.008%	-0.001%	0.595	Marginal
$\epsilon_H = +5\%$	1.051	+0.020%	-0.003%	1.124	Not identifiable
$\Delta r = -20$ mm	1165	+61.7%	+10.1%	1166	Not identifiable
$\Delta r = -10$ mm	571.7	+26.2%	+4.85%	572.1	Not identifiable
$\Delta r = +10$ mm	549.6	-19.5%	-4.47%	549.2	Not identifiable
$\Delta r = +20$ mm	1077	-34.2%	-8.58%	1077	Not identifiable

Radial parameter biases are the linear projection; see the nonlinear check before reading them literally. Pre-amendment: depth $d_{\text{sys}} \leq 0.205$, all Identifiable; radial d_{sys} 98 to 209.

Tolerance (first crossing, - / + side)	$d_{\text{sys}} = 0.5$	$d_{\text{sys}} = 1$	$ \delta\beta/\beta = 1\%$
Water depth ϵ_H	-2.42 / +2.40%	-4.88 / +4.77%	not crossed within $\pm 5\%$ (max 0.021%)
Radial Δr	$\pm 8.94 \mu\text{m}$	$\pm 17.9 \mu\text{m}$	-0.443 / +0.447 mm
pre-amendment	depth none, radial 50 μm	depth none, radial 100 μm	depth none, radial 0.29 mm

The α range of the depth cases. Rescaling H rescales α . The *retained* bins stay inside the validated window [1.7, 13.85] in all four cases (at +5% the retained maximum is 13.807). The wider synthesis support leaves it (support [1.615, 13.157] at -5%, maximum 14.542 at +5%), which is the source of the `computeSensorFRF` warnings during the

tolerance search. The runner flags all four as extrapolated; this is a caveat on the support region, not a failure of the retained features.

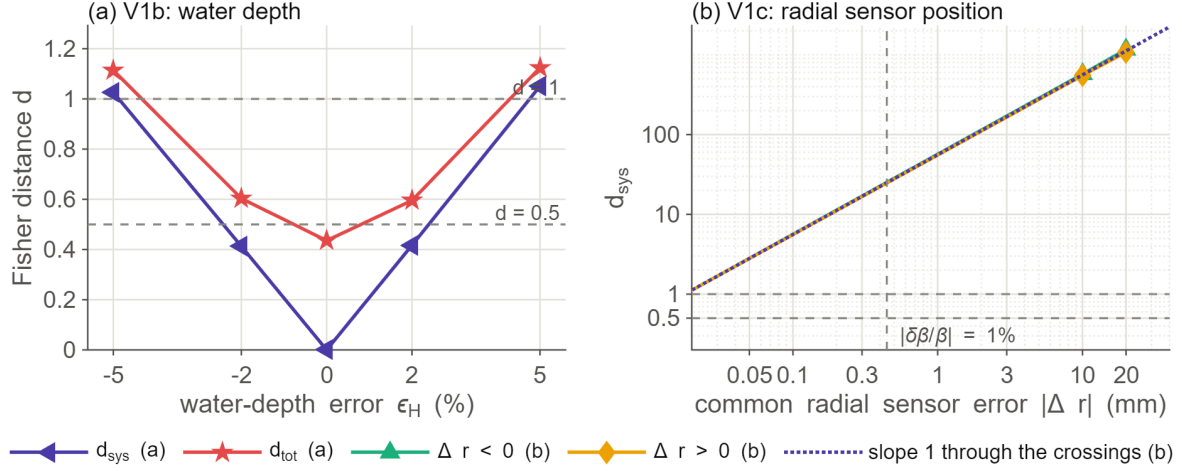


Figure 2. (a) d_{sys} and d_{tot} against the water-depth error (the $\epsilon_H = 0$ point is the matched model: $d_{sys} = 0$, $d_{tot} = d_W$). (b) d_{sys} against the common radial sensor error, both signs, log-log, with the registered ± 10 and ± 20 mm points; dotted: slope 1 through the crossings; dashed vertical: the $|\delta\beta/\beta| = 1\%$ crossing. The $d = 0.5$ and 1 crossings (9 and $18 \mu\text{m}$) lie at the left edge of the axis.

2.4 Nonlinear check

The largest d_{sys} is $\Delta r = -20$ mm. The solver converged in 6 iterations (29 model evaluations, 17 min) to $\delta\beta/\beta = +51.3\%$ and $\delta R/R = +9.08\%$, against the linear totals $+61.7\%$ and $+10.1\%$. The gap to the linear total is $[-133.8, -112.7]\sigma$ ($d = 135$; the 0.2σ reference is not a gate for a diagnostic this far out). The linear projection overstates the stiffness bias by about a fifth at 2 cm. The sign and order of magnitude hold, but the value does not.

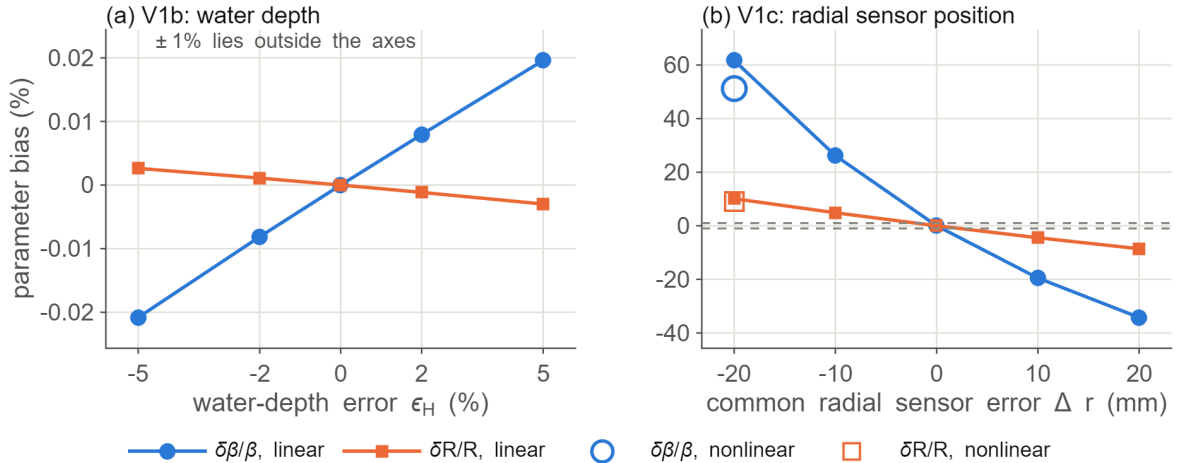


Figure 3. Linearised parameter bias against (a) depth error and (b) radial error, with the nonlinear inversion at $\Delta r = -20$ mm (open markers). In (b) the dashed lines at $\pm 1\%$ sit almost on zero; in (a) they lie outside the axes.

3 Interpretation

- Two different questions now have different answers.** The Fisher-unit criteria ($d = 0.5, 1$) ask whether a model error is visible against the estimator noise. The physical criterion (1% in β) asks whether it matters for the floe property. Before the amendment the two roughly agreed; now $\sigma_{\ln\beta} = 0.05\%$, so any model error above a few hundredths of a percent is visible. This is the same separation R4 found for the draught. The thesis should report both and say which one governs which decision.
- Truncation dominates the statistical error budget.** $d_{sys} \geq 10$ at every floe, 23 to 38 times the Welch bias, and it makes every floe Not identifiable. A1 concluded that σ and the class had converged in the truncation, and that remains true *within the model*. V1a shows that the *features* of $[50 \ 10 \ 10]$ differ from a more converged model by far more than the estimator can resolve. Physically the error is small (at most 0.18% in β), so the production model is adequate for the design studies and the trends. It is not adequate for an inverse that claims 0.05% precision on real data.
- A converged reference was not reached.** The vertical-mode series converges slowly: about 0.008% in β per 20 modes at $M = 210$, with P also unconverged. For real data, either the inverse runs at a high truncation that is checked per floe, or the truncation error is carried as a model-error term in the covariance. Extending the series is outside this project's time.
- Depth: statistically visible, physically negligible.** A 2.4% depth error gives $d_{sys} = 0.5$, but the stiffness moves by only 0.02% at $\pm 5\%$, fifty times inside the 1% tolerance. Depth measured to within $\pm 2\%$ keeps the classes intact. Any reasonable depth measurement satisfies the physical tolerance.

5. **Radial geometry is still the dominant practical problem.** d_{sys} grows linearly with $|\Delta r|$ (slope 1 in Figure 2b, about 56 per mm), symmetric in sign at the crossings (± 8.94 and $\pm 17.9 \mu\text{m}$). This marks it as a genuine small-perturbation result. For scale, $\sigma_{\ln R} = 0.0083\%$ is about 0.06 mm on the basin floe radius, so a common sensor shift of a sixth of the radius uncertainty already biases the estimate by half a standard deviation. Physically, a 1% stiffness bias needs only 0.44 mm, which is the one physical tolerance in V1 that is hard to meet without calibration.
6. **The centimetre-scale numbers are qualitative.** Given the 135 Fisher-unit nonlinear gap, the ± 10 and ± 20 mm parameter biases must not be read literally. They show only that such errors lie far outside the local regime. The quantitative radial results are the small-error crossings at $9 \mu\text{m}$ to 0.44 mm.
7. **Biases add as vectors.** For depth the systematic and Welch biases are now nearly orthogonal ($\sqrt{0.417^2 + 0.436^2} = 0.603$ against $d_{\text{tot}} = 0.595$ at $+2\%$). Adding distances would put the $\pm 2\%$ cases at about 0.85 instead of 0.60. The vector combination fixed in the method still matters at the class boundaries.

Conclusion (thesis wording). Breaking the inverse crime exposed forward-model error as the dominant limitation once the estimator became precise. Against the most converged truncation reachable, [210 20 20], the production model [50 10 10] produced Fisher-scaled systematic biases of $d_{\text{sys}} = 10$ to 29 at the four test floes, making all of them Not identifiable, although the parameter shifts were at most 0.18% in β and 0.12% in R . The reference itself was not converged, so these are lower bounds. Water-depth errors reached $d_{\text{sys}} = 0.5$ at about $\pm 2.4\%$ but moved β by only 0.02% at $\pm 5\%$. Coherent radial sensor-coordinate errors were the most influential: $d_{\text{sys}} = 0.5$ and 1 occurred at about 9 and $18 \mu\text{m}$, and a 1% stiffness bias at about 0.44 mm. Centimetre-scale errors lay far outside the linear regime, as confirmed by the nonlinear inversion. At the precision this design achieves, model and geometric uncertainty, not measurement noise, set what can be inferred from real data.

Limitations.

- The truth truncation is not converged (M and P), so V1a gives lower bounds.
- V1c tests one coherent radial shift common to all sensors, not independent random, angular or GPS-type errors, which may partly average out or be absorbed by R .
- Biases are linear projections, checked nonlinearly only at the extreme radial case.
- Depth and radial cases are at p_0 only; truncation covers four floes.
- Beyond the deliberate mismatch, truth and inverse share the EMM physics, so other physical model error (plate theory, draught model, floe shape) is outside V1.

Requirement for the experiment and the thesis

Truncation: [50 10 10] remains adequate for the σ -based design and domain trends. The bias-based classes of D1, D2, A1, R1 to R4 are within-model and exclude a truncation bias that is at least 10 to 30 times larger. For real data, invert at a truncation checked for the floe at hand (at least [210 20 20]) or carry the truncation error as model uncertainty.

Depth: within $\pm 2\%$ keeps the statistical classes; within $\pm 5\%$ is physically sufficient (0.02% in β).

Geometry: the radial coordinates of the sensors relative to the assumed floe centre must be known to better than 0.4 mm for a 1% stiffness bias, and to about $10 \mu\text{m}$ to stay statistically invisible. In practice they must be calibrated or jointly estimated. Remedies: geometric calibration, photogrammetry, or adding position parameters to the inverse (to be planned with E0).

Output (Results/V1/)	Contents
V1.20261009.105051.mat / _summary.txt	results and config; headline numbers
V1.truncation.csv, V1.mismatch.csv	V1a per floe; V1b and V1c per case
V1.tolerances.csv, V1.nonlinear.csv	bisected crossings; nonlinear check
diagnostics/truncationLadder.20261009.101105.mat	stage 0 ladder, large- kR and P checks, preview
figures/V1_fig1 to V1_fig3 (.pdf/.png)	Figures 1 to 3 (plotV1ModelMismatch)
Superseded	V1.20261004.121316, truncationLadder.20261004.113129 (pre-amendment)